1. UML Class Diagram

The diagram consists of four main classes: Guest, Room, Booking, and Payment. These classes represent the essential information and activities involved in managing hotel reservations.

2. Classes, Attributes, and Methods

2.1 Guest Class

Description: The Guest class represents a person who makes a hotel reservation. It stores the guest's personal information and supports basic guest management activities.

Attributes:

- guestId: int
- fullName: String
- phone: String

Methods:

- makeBooking(): void
- updateDetails(): void

2.2 Room Class

Description: The Room class represents a hotel room. It stores room information and provides operations for checking availability and updating room status.

Attributes:

- roomNumber: int
- roomType: String
- pricePerNight: double
- status: String

Methods:

- checkAvailability(): boolean

- updateStatus(): void

2.3 Booking Class

Description: The Booking class represents a reservation made by a guest for a hotel room. It stores booking details and supports booking management.

Attributes:

- bookingId: int
- checkInDate: Date
- checkOutDate: Date
- status: String

Methods:

- createBooking(): void
- cancelBooking(): void
- calculateTotal(): double

2.4 Payment Class

Description: The Payment class represents a payment associated with a hotel booking. It stores payment details and provides operations for recording payment information and issuing receipts.

Attributes:

- paymentId: int
- amount: double
- paymentDate: Date
- paymentMethod: String

Methods:

- processPayment(): void
- issueReceipt(): void

3. Relationships Between Classes and Multiplicities

The Hotel Booking System contains three main relationships.

3.1 Guest and Booking

A Guest can make zero or many Bookings over time. Each Booking belongs to exactly one Guest.

Relationship type: Association

Multiplicity:

- Guest end: 1
- Booking end: 0..*

3.2 Room and Booking

A Room can be associated with zero or many Bookings over time. Each Booking refers to exactly one Room.

Relationship type: Association

Multiplicity:

- Room end: 1
- Booking end: 0..*

This relationship represents the history of reservations for a room. The system must prevent conflicting reservations for the same room during overlapping dates.

3.3 Booking and Payment

A Booking can have zero or one Payment in this simplified model. Each Payment belongs to exactly one Booking.

Relationship type: Association

Multiplicity:

- Booking end: 1
- Payment end: 0..1

A booking may have no payment recorded yet, while a completed payment is associated with one booking.

4. Brief Explanation of the Relationships

The association between Guest and Booking connects guests with their reservations. It allows the system to identify who made each booking and to maintain a guest's booking history.

The association between Room and Booking connects reservations to the rooms being reserved. It allows the system to identify which room is assigned to each booking and to track reservations for individual rooms over time.

The association between Booking and Payment connects a reservation with its payment record. It allows the system to track whether payment information has been recorded for a particular booking.